

Comparing Louvain and Leiden community detection performance across network topologies

Laylaa Jorge Abrahams, Khethani Mugeru

Supervisor: Mr. Stefan Britz



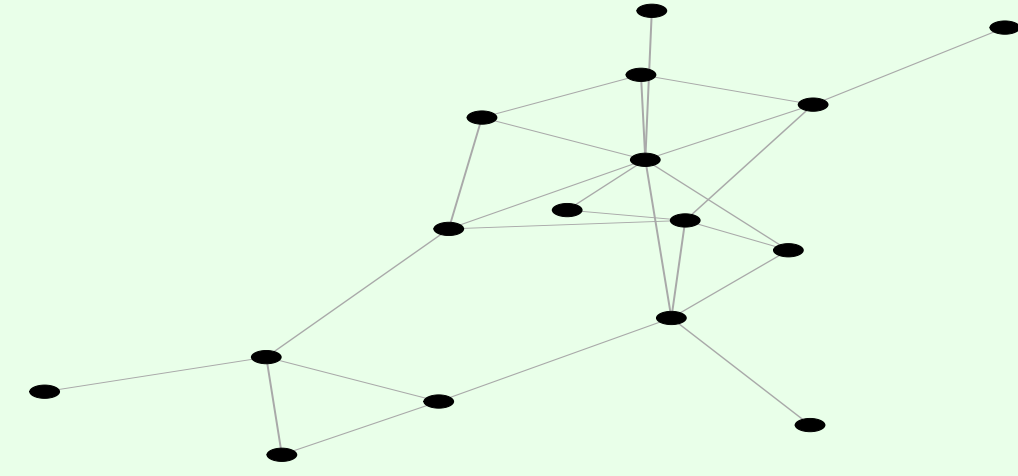
Introduction

Networks can be analysed to reveal natural groupings of nodes, known as communities, based on their connectivity. This project compares the performance of two quality-maximizing community detection algorithms, Louvain and Leiden, on three network topologies: Erdős-Rényi (random), Watts-Strogatz (small-world), and Barabási-Albert (scale-free). Networks of different sizes and mean degrees are generated, and the algorithms' resolution parameter is varied. The performance is compared using metrics such as modularity, conductance and stability.

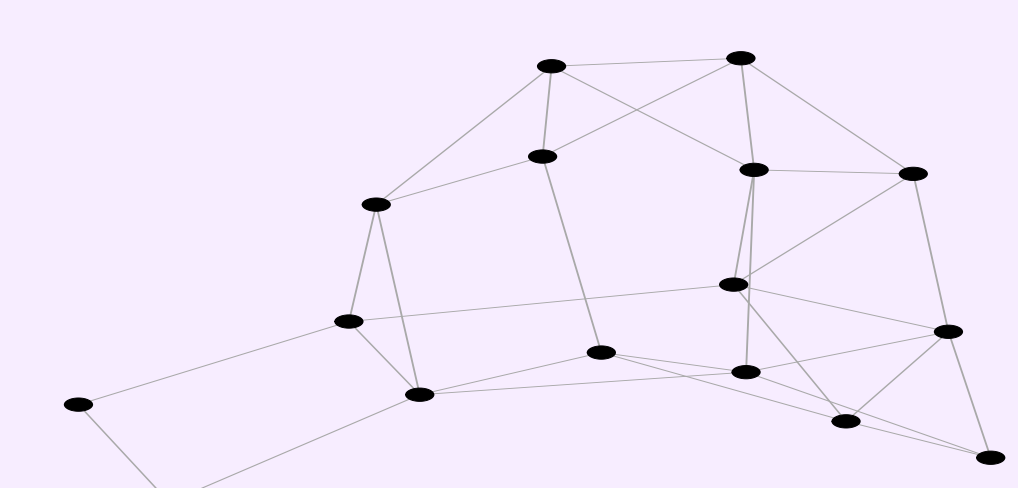
Network Topologies

A network consists of nodes and edges, where the edges connect the nodes. Each network topology has different structural properties, which define how the edges are arranged. This study focuses on three network topologies.

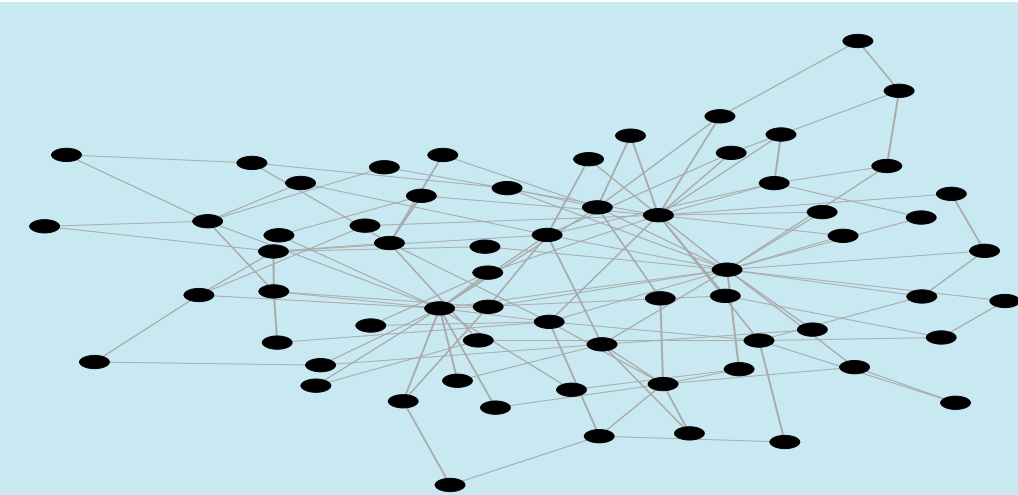
Random: These networks start with isolated nodes; each pair of nodes connects independently with probability p .



Small-world: These networks begin with a regular lattice. The edges of this lattice are rewired at random, producing high clustering and short path lengths.

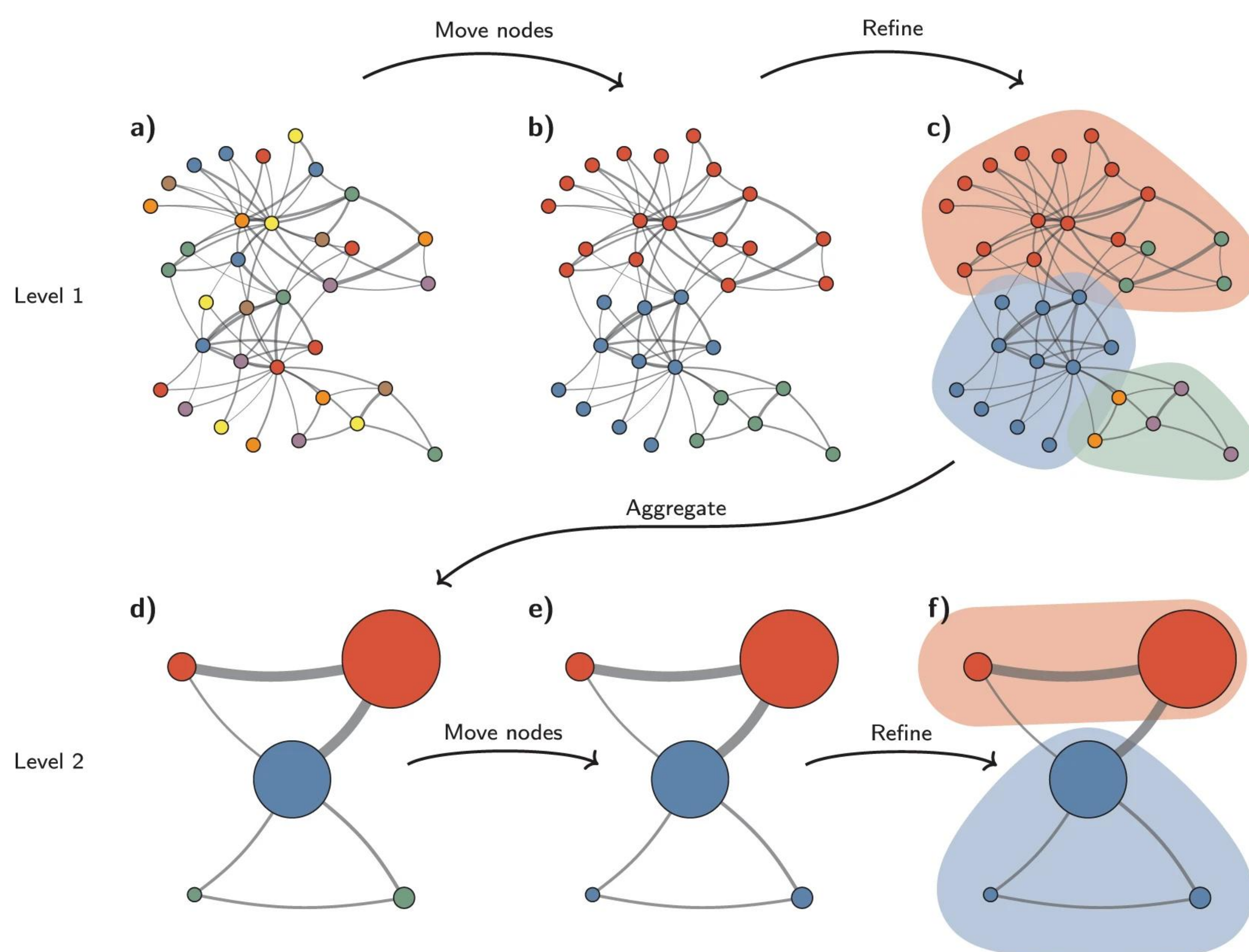


Scale-free: These networks are generated via preferential attachment, meaning new nodes are more likely to connect to highly connected nodes, producing hubs and heavy-tailed degree distributions.

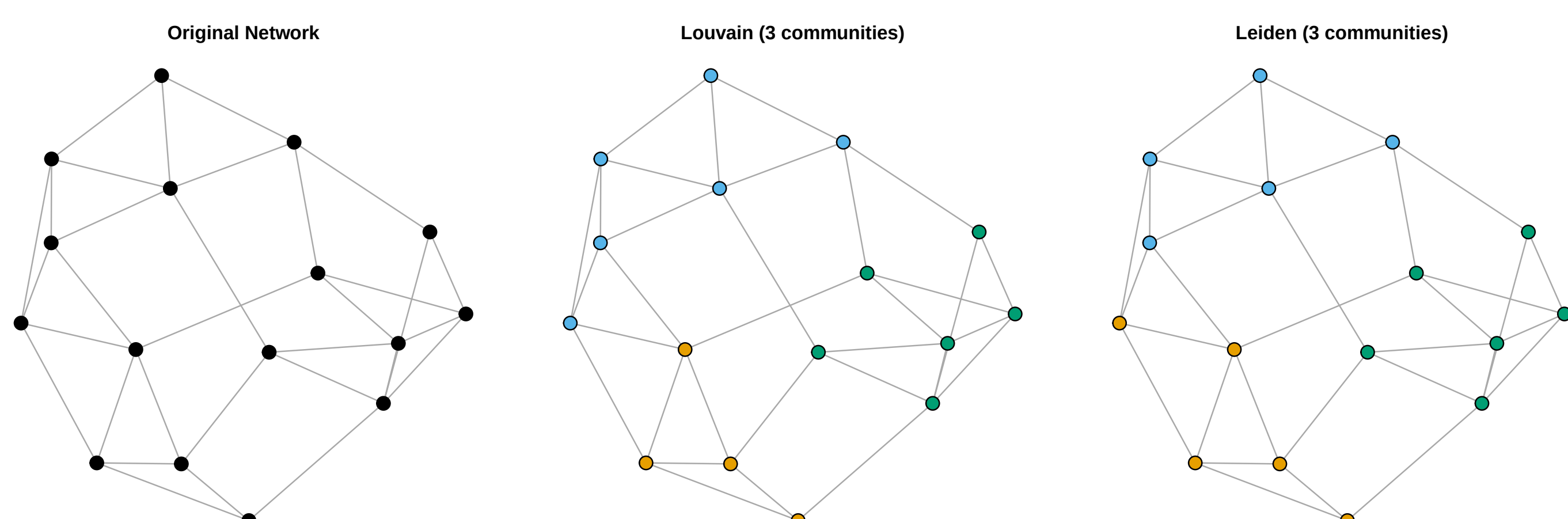


Community Detection

Community detection finds groups of nodes that are more densely connected to each other than to the rest of the network. We use two community detection algorithms: Louvain and Leiden. The Louvain algorithm optimizes modularity using a local moving and aggregation phase but can produce communities that are not internally well connected. The Leiden algorithm addresses this by adding a refinement phase that splits poorly connected communities. The figure below from Traag, Waltman, and Eck (2019) shows the Leiden algorithm phases:



Example: A small-world network shown before community detection, and after partitioning with Louvain and Leiden.



Performance metrics

Modularity is defined as the quality of a particular division of a network.

$$\mathcal{H} = \frac{1}{2m} \sum_c \left[E_c - \gamma \frac{k_c^2}{2m} \right]$$

Conductance assesses how well separated a community is from the rest of the network.

$$\phi(G) = \min_{S \subset V} \phi(S) = \min_{S \subset V} \frac{\text{vol}(E(S))}{\min\{\text{vol}(S), \text{vol}(\bar{S})\}}$$

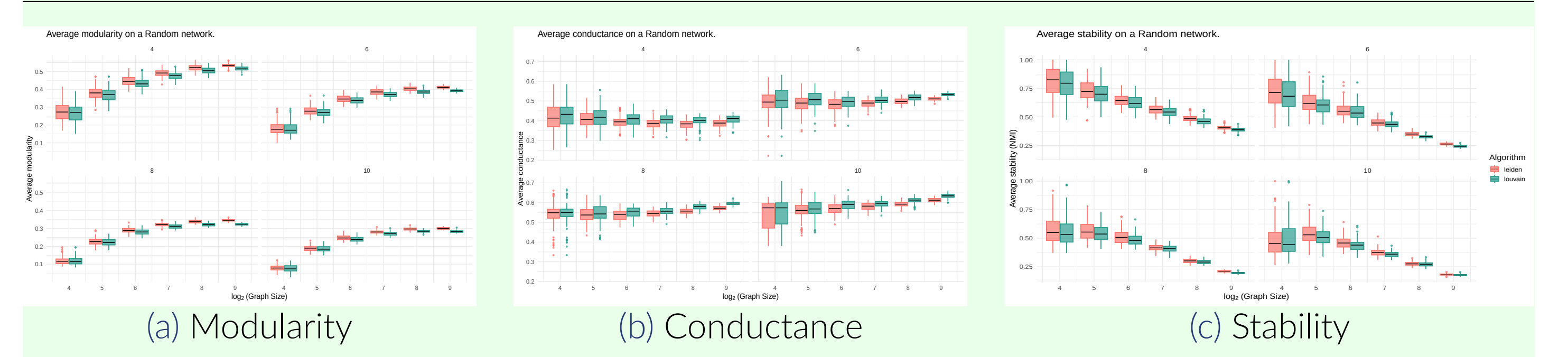
Stability refers to how consistently an algorithm partitions a network when it is run multiple times under the same settings.

$$\text{NMI}(X, Y) = \frac{2I(X; Y)}{H(X) + H(Y)}$$

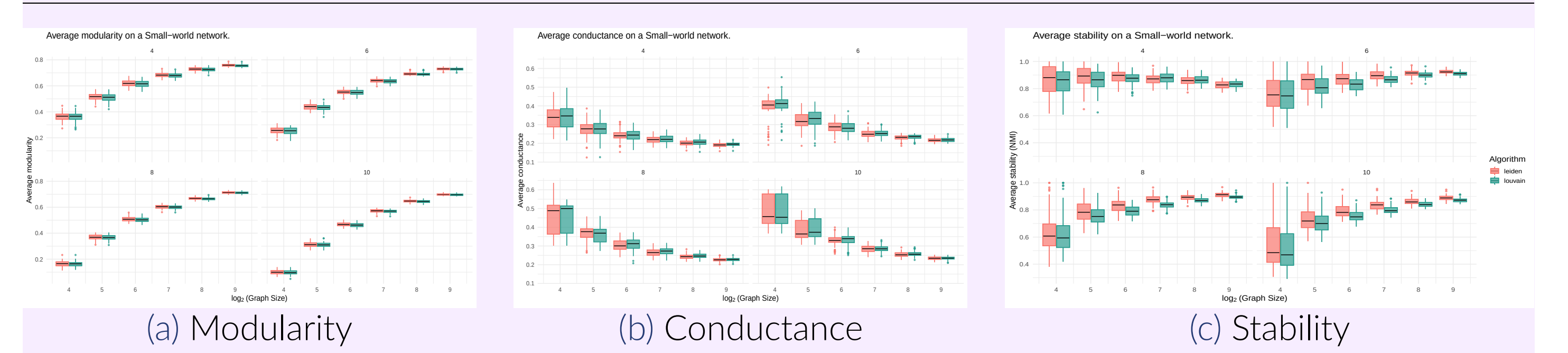
Experimental Design

We analysed networks of sizes $n \in 16, 32, 64, 128, 512$, mean degrees 4, 6, 8, 10, and resolution $\gamma \in [0.5, 5.0]$. Stage 1 varied network size and mean degree at resolution (γ) = 1. Stage 2 varied the resolution (γ) and network size at a constant mean degree. Results show the average metrics over 100 runs per configuration, comparing the performance of the Louvain (blue) and Leiden (orange) algorithms.

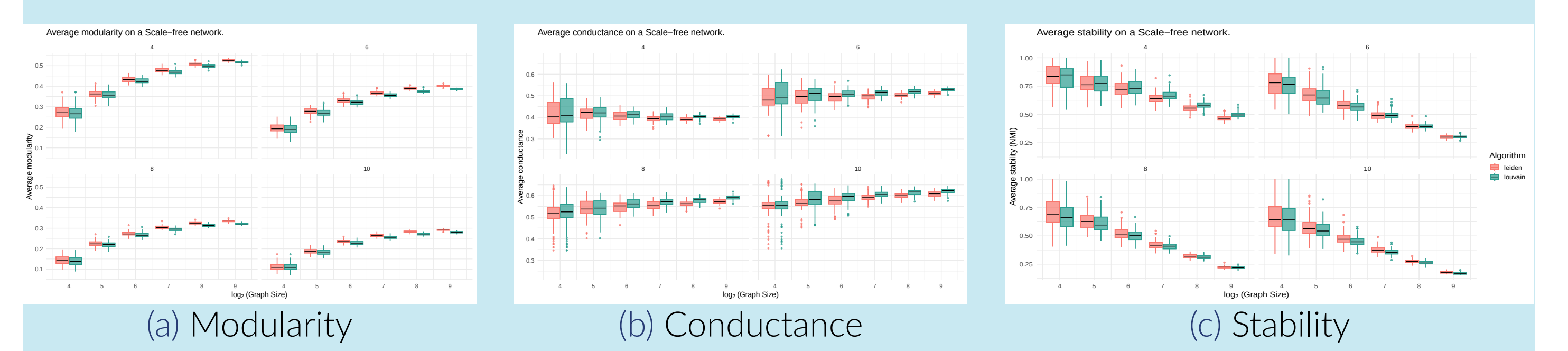
Random Network Results



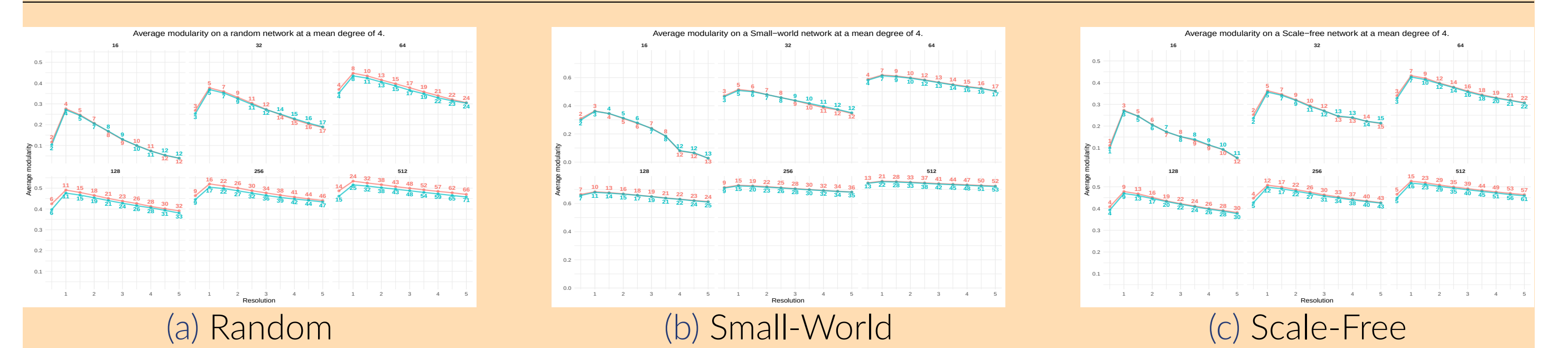
Small-World Network Results



Scale-Free Network Results



Modularity Comparison (Mean degree = 4)



The modularity results presented above correspond to Stage 2 of the analysis, where the resolution parameter and network size was varied at a constant mean degree of 4 across each network topology. In all three network topologies the modularity decreases as resolution increases.

Findings and Conclusion

- Leiden generally outperforms Louvain, achieving higher modularity, lower conductance, and greater stability, although the differences are often minor.
- There is one exception: In scale-free networks with a mean degree of four and resolution one, Louvain produces slightly more stable partitions.
- Network topology plays a defining role: Small-world networks show strong community structure, scale-free networks display moderate structure shaped by hub nodes, and random networks lack well-defined community structures.
- These results highlight that both the choice of algorithm and the underlying network topology substantially influence community detection outcomes.

References

Traag, V.A., L. Waltman, and N.J. Van Eck (2019). "From Louvain to Leiden: guaranteeing well-connected communities". In: *Scientific Reports* 9.1, p. 5233.